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Paper 1: Assessment and Benchmarking of Model Informed Approaches in Drug Development for

Bibliographic Information

- **Title**: Assessment and Benchmarking of Model Informed Approaches in Drug Development for Hemoglobinopathies: A Review of Scientific Advices From January 2000 to December 2024
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- **Journal**: CPT: Pharmacometrics & Systems Pharmacology, 2026
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Quick Take

A comprehensive review of 25 years of EMA Scientific Advice for sickle cell disease and thalassemia, benchmarking model-informed approaches against standard statistical methods. Essential reading for regulatory scientists and modelers working in rare diseases to understand how ICH M15 principles are applied to model acceptance.

Executive Summary

This systematic review analyzes the uptake and regulatory endorsement of Model-Informed Drug Development (MIDD) in hemoglobinopathies by examining 76 EMA Scientific Advice procedures from 2000 to 2024. Utilizing a 'Question-Centric Approach' and the ICH M15 credibility framework, the authors constructed a repository of regulatory questions to benchmark M&S methods against traditional approaches. The study reveals that while M&S (primarily PopPK and Exposure-Response) is increasingly used for dose selection and pediatric extrapolation, it remains underutilized, appearing in only ~5-6% of total regulatory questions. The authors argue that widespread adoption requires transparent mapping of 'Model Impact' and 'Model Risk' to regulatory questions, positioning the ICH M15 framework as a critical tool for aligning sponsor strategies with regulatory expectations.

⚡ Methodological Snapshot

The authors conducted a retrospective analysis of EMA Scientific Advice (SA) and Protocol Assistance documents using the EMA IRIS platform and Scientific Explorer. They applied a 'Question-Centric Approach' to translate sponsor queries into standardized regulatory questions classified by hierarchy (Benefit-Risk > Efficacy/Safety/PK > Data/Methods). The study benchmarked M&S approaches against alternative methods (e.g., standard statistical analysis) and applied the ICH M15 credibility assessment framework to evaluate regulatory feedback.

🏗 Structural Model Breakdown

This is a meta-research paper and does not propose a new mathematical model. However, it establishes a 'Repository Structure' for regulatory questions: 1. Root: Benefit-Risk Assessment. 2.

Layers: Characterization of Efficacy, Safety, and PK. 3. Sub-layers: Molecular, Cellular, Tissular, Individual, and Population levels. 4. Benchmarking: Mapping specific M&S methods (e.g., PopPK) and alternative methods (e.g., Allometric Scaling) to these standardized questions.

Detailed Methodological Analysis

Modeling Approach

Systematic review and meta-analysis of regulatory documents. Application of the ICH M15 credibility framework to assess 'Model Impact' and 'Model Risk'. Classification of questions using a hierarchical Benefit-Risk structure.

Data Sources

45 Scientific Advices for Sickle Cell Disease and 31 for Thalassemia (Total: 76 procedures) issued by the EMA CHMP between January 2000 and December 2024.

Estimation Methods

N/A (Qualitative research). The review notes that applicants primarily used Population PK (PopPK) and Exposure-Response (E-R) modeling, with occasional use of PBPK and MMRM.

Model Evaluation

The authors evaluated the 'acceptability' of methods based on CHMP feedback keywords (e.g., 'supported', 'agreed', 'questionable'). M&S methods were benchmarked against standard statistical approaches (e.g., ANOVA, allometric scaling) used for identical regulatory questions.

Covariate Analysis

N/A (Review paper). The study notes that covariate analysis in applicant models was frequently used to support extrapolation to specific populations (pediatrics, organ impairment).

Scientific Context & Motivation

Hemoglobinopathies like Sickle Cell Disease (SCD) and Thalassemia present significant drug development challenges, particularly regarding recruitment and pediatric extrapolation. While MIDD is touted as a solution to improve efficiency, its actual penetration into regulatory decision-making for these indications has not been quantified. This study addresses the gap between the theoretical promise of MIDD and its practical application in regulatory interactions, specifically within the European regulatory landscape.

Statistical Rigor Assessment

The study employs a structured, reproducible search strategy using defined keywords and inclusion criteria within the EMA database. The classification of questions was standardized to minimize subjectivity. Descriptive statistics are provided regarding the frequency of M&S usage across different development phases.

📊 Key Findings

M&S was discussed in 33% of SCD and 20% of Thalassemia advices, but represented only ~6% of total specific questions asked. PopPK and E-R analyses were the dominant methodologies. M&S was most frequently applied in Phase 2 and 3 to support dose selection, study design optimization, and pediatric extrapolation. Benchmarking revealed that M&S is often proposed as an alternative to standard statistical analysis (SSA) for questions regarding dose justification and extrapolation; for instance, PopPK/PD vs. NOAEL for safety characterization, or PopPK vs. weight-based allometric scaling for dosing.

💡 Clinical & Regulatory Implications

The findings highlight that M&S is a viable, though underused, pathway for justifying dosing in pediatric hemoglobinopathy populations, potentially reducing the need for extensive clinical trials in vulnerable groups. The study emphasizes that regulatory acceptance is contingent on clearly defining the 'context of use'—specifically, how the model impacts the benefit-risk assessment.

Strengths & Limitations

Strengths

- Access to confidential EMA Scientific Advice documents provides a unique, real-world view of regulatory thinking.
- First application of the ICH M15 framework to benchmark M&S in this therapeutic area.
- Creation of a structured repository of regulatory questions that facilitates objective comparison of methods.

Limitations (Acknowledged by Authors)

- Analysis is restricted to EMA procedures; FDA or other agency perspectives are not included.
- Retrospective nature limits the analysis to questions actually asked by sponsors, potentially missing missed opportunities for M&S.

Limitations (Expert Review)

- The specific technical reasons for model rejection (e.g., specific shrinkage values, diagnostic failures) are not detailed, likely due to the high-level nature of SA letters.
- Does not quantify the eventual success (approval) of the programs that used M&S vs. those that did not.

Generalizability

High relevance for rare disease drug development in Europe; the principles of the Question-Centric Approach and ICH M15 application are transferable to other therapeutic areas.

Figures & Tables

- **Figure 1**: Flowchart of the methodology: Data searching -> Repository building -> ICH M15 implementation -> Benchmarking.
  - *Significance*: Outlines the systematic review process and how the 'Question-Centric Approach' was operationalized.
- **Figure 3**: Hierarchical structure of the repository, organizing questions from Benefit-Risk down to specific data/methods.
  - *Significance*: Visualizes the conceptual framework used to classify regulatory questions, showing how PK, Efficacy, and Safety feed into the central Benefit-Risk assessment.
- **Table 1**: Frequency of M&S usage across development stages (Non-clinical to Phase 3) for SCD and Thalassemia.
  - *Significance*: Demonstrates that M&S is predominantly used in late-stage development (Phase 2/3) for these indications, rather than early translational phases.
- **Table 2**: Benchmarking of M&S models against alternative analysis methods for specific regulatory questions.
  - *Significance*: The most practical element of the paper; it directly compares M&S approaches (e.g., PopPK, E-R) with standard alternatives (e.g., SSA, NOAEL) and assigns a 'Model Impact' level.

Supplementary Materials

Table S1 correlates CHMP comments/recommendations to key assessment elements of the ICH M15 credibility framework, providing illustrative examples of regulatory feedback.

Future Directions

The authors propose the ERAMET project will expand on this by establishing a quantitative framework to link 'Model Risk' and 'Model Impact' to specific acceptance criteria. Future work should focus on creating an open-access framework of standards that make a method acceptable for regulators.

Expert Commentary

This paper is a timely operationalization of the ICH M15 guideline. It shifts the conversation from 'is the model valid?' to 'is the model fit for the specific regulatory question?'. The benchmarking table (Table 2) is particularly instructive for pharmacometricians, as it explicitly maps M&S tasks to their 'Standard Statistical' counterparts (e.g., PopPK vs. Allometry). This helps in strategic planning: if you choose M&S, you must demonstrate it offers a better benefit/risk characterization than the standard alternative. The low utilization rate (5-6% of questions) suggests a significant educational gap or risk aversion in industry that frameworks like this aim to correct.

Bottom Line

For pharmacometricians working in rare diseases, this paper provides a roadmap for regulatory engagement. It demonstrates that while regulators accept M&S for critical decisions like dose selection and pediatric extrapolation, success depends on explicitly mapping the model to a regulatory question and defining its impact level per ICH M15. M&S is not just a technical exercise but a strategic tool to be benchmarked against standard statistical methods.

📏 Figures

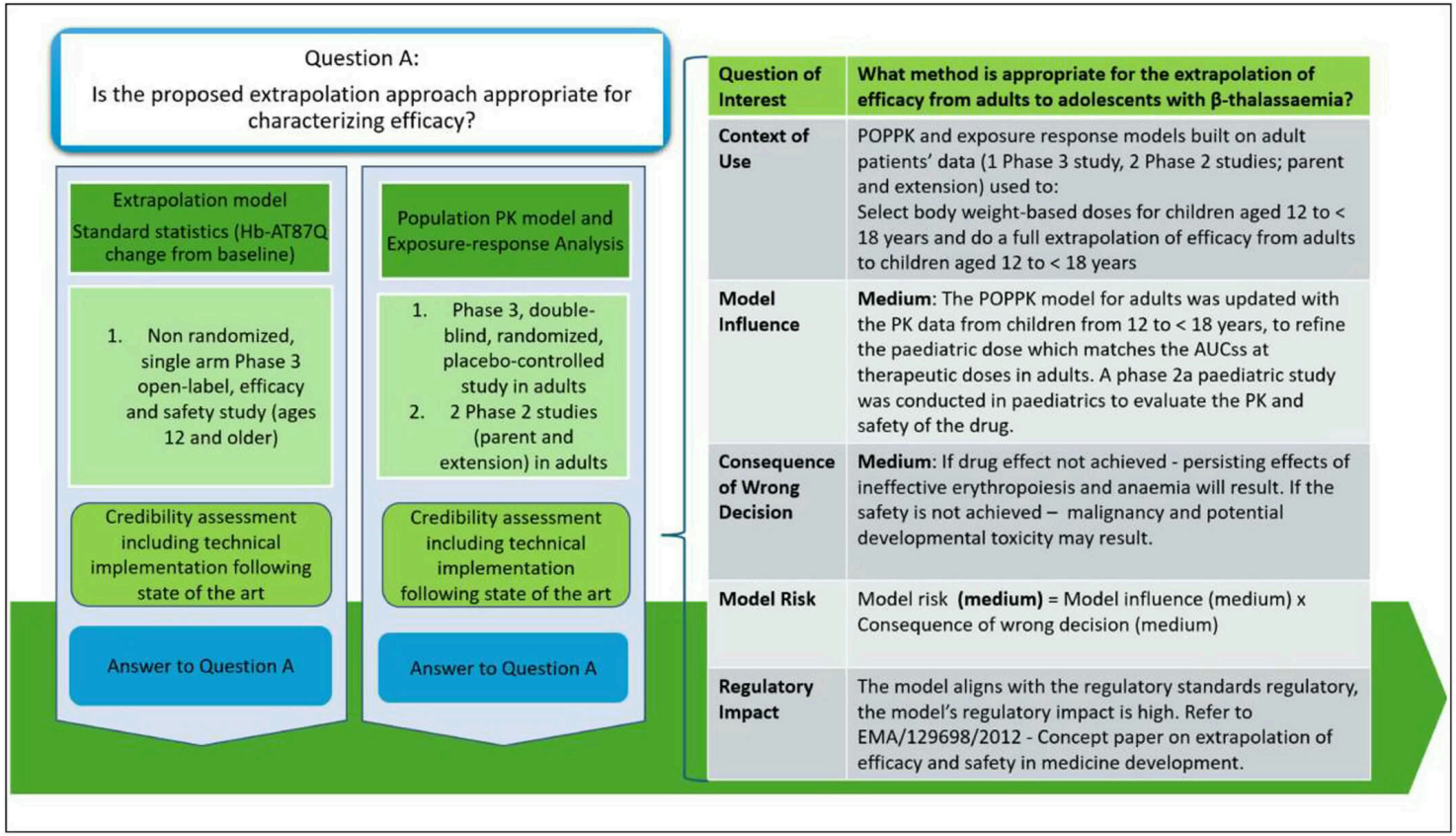

Figure 1: FIGURE 5: Credibility assessment of a model and benchmarking against alternative statistical analysis for addressing a drug development question related to extrapolation o